SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks.* *(BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I HARINI G/192524011 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Harinigopi
Signature of the student:

Annexure A

Short Answer Test

1. A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer.

Given: File size = 150 MB, segment size = 1500 bytes, Data Link overhead = 40 bytes/frame, bandwidth = 50 Mbps.

Calculations

Number of segments = 150,000,000 ÷ 1500 = 100,000 segments

Number of frames = 100,000 frames (1 segment per frame)

Total overhead = 100,000 × 40 = 4,000,000 bytes (4 MB)

Total data transmitted = 150,000,000 + 4,000,000 = 154,000,000 bytes

Transmission time = (154,000,000 × 8) ÷ 50,000,000 = 24.64 seconds

2.A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given

- Window size = **6 frames**
- Each frame = **1024 bytes**
- Total frames = **48**

$$\text{Number of windows} = \frac{48}{6} = 8$$

Transmission windows = 8

- 1 frame is lost in each window.
- Total windows = 8

$$\text{Retransmissions} = 8 \times 1 = 8$$

Total retransmissions = 8

Final Answer

- **Transmission windows: 8**
- **Retransmissions: 8 frames**

FlowControl:

Flow control controls the speed of data transmission so the receiver is not overloaded, improving communication efficiency.

3.A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Given

- IP packet size = **12,000 bytes**
- MTU = **1500 bytes**
- IP header = **20 bytes**

$$1500 - 20 = 1480 \text{ bytes}$$

Data per fragment = 1480 bytes

$$\frac{12000}{1480} = 8.1$$

Round up to the next whole number:

Number of fragments = 9

$$9 \times 20 = 180 \text{ bytes}$$

Total header overhead = 180 bytes

Final Answer

- **Fragments created: 9**
- **Total header overhead: 180 bytes**

NetworkLayer:

The Network Layer fragments large packets, assigns fragment information, and reassembles them at the destination to ensure successful delivery.

4.A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given

- Sliding window size = **6 frames**
- Each frame = **1024 bytes**
- Total frames = **48**

$$\text{Number of windows} = \frac{48}{6} = 8$$

Transmission windows = 8

- 1 frame is lost in each window.
- Total windows = 8

$$\text{Retransmissions} = 8 \times 1 = 8$$

Total retransmissions = 8 frames

Final Answer

- **Transmission windows: 8**
- **Retransmissions: 8 frames**

FlowControl:

Flow control prevents the sender from sending data faster than the receiver can handle, reducing data loss and improving communication efficiency.

5.A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs after 390 MB has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Given

- Video file size = **600 MB**
- Checkpoint after every **60 MB**
- Failure occurs after **390 MB**

Step 1: Calculate the number of checkpoints before failure

$$\frac{390}{60} = 6 \text{ checkpoints (at 60, 120, 180, 240, 300, 360 MB)}$$

Checkpoints before failure = 6

Last checkpoint = **360 MB**

Failure occurs at **390 MB**

$$390 - 360 = 30 \text{ MB}$$

Data to be retransmitted = 30 MB

Final Answer

- **Checkpoints created before failure: 6**
- **Data to be retransmitted: 30 MB**

SessionLayer:

The Session Layer uses synchronization checkpoints so that after a failure, transmission resumes from the last checkpoint instead of restarting from the beginning, saving time and bandwidth.

6.Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Given

- Original file size = **900 MB**
- Compression = **40%**
- Encryption overhead = **5%** of compressed size

$$900 - (40\% \times 900) = 900 - 360 = 540 \text{ MB}$$

Compressed file size = 540 MB

$$540 + (5\% \times 540) = 540 + 27 = 567 \text{ MB}$$

Encrypted file size = 567 MB

$$\frac{900 - 567}{900} \times 100 = 37\%$$

Total percentage reduction = 37%

Final Answer

- **Compressed file size: 540 MB**
- **Encrypted file size: 567 MB**
- **Total reduction: 37%**

PresentationLayer:

The Presentation Layer compresses data to reduce transmission size and encrypts it to provide security during transmission.

7. An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The Application Layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users

Given

- Data size = **3 GB = 3000 MB**

- Request size = **3 MB**
- Throughput = **120 Mbps**

Calculate the number of requests

$$\frac{3000}{3} = 1000$$

Total requests = 1000

Calculate the transmission time

Convert data to bits:

$$3000 \times 8 = 24000 \text{ Mb}$$

$$\text{Transmission time} = \frac{24000}{120} = 200 \text{ seconds}$$

Transmission time = 200 seconds

Final Answer

- **Number of requests: 1000**
- **Transmission time: 200 seconds**

ApplicationLayer:

It provides services such as file transfer, error handling, and user authentication to ensure reliable communication between users and applications.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms, and the transmission delay is 12 ms. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Given

- Application Layer delay = **4 ms**
- Transport Layer delay = **3 ms**
- Network Layer delay = **5 ms**
- Data Link Layer delay = **2 ms**
- Physical Layer delay = **1 ms**
- Propagation delay = **18 ms**

- Transmission delay = **12 ms**

Calculate total processing delay

$$4 + 3 + 5 + 2 + 1 = 15 \text{ ms}$$

Processing delay = 15 ms

Calculate total end-to-end delay

$$15 + 18 + 12 = 45 \text{ ms}$$

Total end-to-end delay = 45 ms

Final Answer

- **Processing delay: 15 ms**
- **Total end-to-end delay: 45 ms**

OSI Layer Contribution:

- **Application Layer:** Provides network services to applications.
- **Transport Layer:** Ensures reliable data delivery.
- **Network Layer:** Routes packets to the destination.
- **Data Link Layer:** Performs framing and error detection.
- **Physical Layer:** Transmits data as electrical or optical signals.

9. A network transfers 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Given

- Useful data = 80 MB = 80,000,000 bytes
- Frame size = 1600 bytes
- Protocol overhead = 80 bytes/frame

Step 1: Calculate useful data per frame

$$1600 - 80 = 1520 \text{ bytes}$$

Useful data per frame = 1520 bytes

Step 2: Calculate the number of frames

$$\frac{80{,}000{,}000}{1520} = 52{,}631.58$$

Round up to the next whole number:

Number of frames = 52,632

Step 3: Calculate the total overhead

$$52{,}632 \times 80 = 4{,}210{,}560 \text{ bytes}$$

Total overhead = 4,210,560 bytes

Step 4: Calculate transmission efficiency

Total transmitted data:

$$80{,}000{,}000 + 4{,}210{,}560 = 84{,}210{,}560 \text{ bytes}$$

$$\text{Efficiency} = \frac{80{,}000{,}000}{84{,}210{,}560} \times 100$$

$$= 94.99\% \approx 95\%$$

Transmission efficiency = 95%

Final Answer

- Number of frames: 52,632
- Total overhead: 4,210,560 bytes
- Transmission efficiency: 95%

Protocol Overhead:

Protocol overhead uses extra bytes for headers and trailers, reducing efficiency but ensuring reliable communication, addressing, and error detection.

10. A hospital transfers a patient's medical record of 240 MB through an IoT-enabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Given

- Original file size = **240 MB = 240,000,000 bytes**

- Compression = **30%**
- Segment size = **1400 bytes**
- Data Link overhead = **60 bytes/frame**

Step 1: Calculate the compressed file size

$$240 - (30\% \times 240) = 240 - 72 = 168 \text{ MB}$$

$$168 \text{ MB} = 168{,}000{,}000 \text{ bytes}$$

Compressed file size = 168 MB

Step 2: Calculate the number of segments

$$\frac{168{,}000{,}000}{1400} = 120{,}000$$

Number of segments = 120,000

Step 3: Calculate the total protocol overhead

$$120{,}000 \times 60 = 7{,}200{,}000 \text{ bytes}$$

Total protocol overhead = 7,200,000 bytes (7.2 MB)

Step 4: Calculate the final data transmitted

$$168{,}000{,}000 + 7{,}200{,}000 = 175{,}200{,}000 \text{ bytes}$$

$$= 175.2 \text{ MB}$$

Final data transmitted = 175.2 MB

Final Answer

- **Compressed file size: 168 MB**
- **Number of segments: 120,000**
- **Protocol overhead: 7.2 MB**
- **Final data transmitted: 175.2 MB**

OSI Layers:

The **Presentation Layer** compresses data, the **Transport Layer** divides it into segments for reliable delivery, the **Data Link Layer** adds headers for error detection, and the **Physical Layer** transmits the data over the network, ensuring secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
		and coherence			
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand